# NIKHIL REDDY LEVAKU

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## PROFESSIONAL SUMMARY

Software Development Engineer with production experience building **100+ RESTful APIs** using Core Java, Spring Boot, and PostgreSQL across the full **software development lifecycle (SDLC)**. Strong foundation in **data structures & algorithms**, OOP, **microservices architecture**, and event-driven systems (Kafka/RabbitMQ). Hands-on experience in concurrency-safe transactional workflows, database optimization, GitHub Actions CI, and **Agile** delivery.

## TECHNICAL SKILLS

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|---|---|
| **Languages & Frameworks** | Java, SQL, Spring Boot, Spring Security, Hibernate/JPA |
| **Databases & Cache** | PostgreSQL, MySQL, Redis, Flyway |
| **Messaging & Systems** | Apache **Kafka**, RabbitMQ, Microservices, Event-Driven Design, RESTful API Design |
| **Testing & DevOps** | JUnit 5, Mockito, Postman, Swagger / OpenAPI, Git, GitHub Actions (CI), Jenkins, SonarQube, SonarScanner, Docker, Docker Compose, Kubernetes, Helm 3, Nginx, Terraform (IaC Modules), Ansible, Bash/Shell Scripting, AWS (EC2, VPC, S3, RDS, CloudWatch), Linux, Gradle, Maven |
| **Concepts** | Data Structures & Algorithms, Concurrency, OOP, System Design, SDLC, Agile / Scrum |

## PROFESSIONAL EXPERIENCE

**DeepLure Research Pvt Ltd — Backend Engineer** **May 2026 – Aug 2026**

Java · Spring Boot · Spring Security · Microservices · PostgreSQL · Redis · RabbitMQ · Infisical · AWS S3 · Docker

- Analyzed business requirements and shipped **100+ production REST APIs** across B2B and on-demand platforms, building configurable backend components for authentication, orders, payments, and admin modules.
- Implemented **JWT** authentication and **role-based access control (RBAC)**, with OTP verification over SMS and email.
- Integrated the **Razorpay** payment gateway (orders, wallets, refunds), applying idempotency keys and concurrency locking to prevent double-charges and race conditions under concurrent requests.
- Used **RabbitMQ** for asynchronous inter-service notification workflows; managed application secrets with **Infisical**.
- Created **AWS S3** buckets for application media storage.
- Engineered **Redis-backed asynchronous workers** for notifications, payment reconciliation, and booking expiry, and reduced database load **~40%** through Redis caching and query optimization.
- Modeled normalized **PostgreSQL** schemas with Hibernate/JPA and Flyway; aligned API contracts with frontend teams via Swagger/OpenAPI; wrote **JUnit 5** and **Mockito** tests; contributed in Agile sprints.

**SmartBridge Educational Services (APSCHE) — Data Analytics Intern** **May 2025 – Jul 2025**

- Cleaned sales datasets with SQL joins and aggregates; built 3 Tableau dashboards for production and inventory review.

## PROJECTS

**Flight Booking System — Distributed Microservices Platform** **Aug 2025 – Jan 2026**

Java · Spring Boot · Microservices · PostgreSQL · Redis · Apache Kafka · Docker · Nginx | GitHub

- Refactored a monolithic Spring Boot backend into **4 independently deployable microservices** (Auth, Flight, Booking, Notification), each with its own **PostgreSQL** database and versioned **Flyway** migrations.
- Implemented inter-service REST communication — the Booking Service calls the Flight Service to reserve seats — propagating the user's **JWT** so each service independently authenticates and enforces RBAC (USER/ADMIN).
- Prevented overbooking under concurrent requests with transactional seat reservation using row-level **pessimistic write locking** (**PESSIMISTIC_WRITE**) inside **@Transactional** methods; added seat rollback logic.
- Built an **event-driven** flow with **Apache Kafka**: Booking Service publishes a booking-created event, consumed asynchronously by a dedicated Notification Service, decoupling notifications from the booking transaction.
- Added **Redis caching** in the Flight Service, request validation, centralized exception handling, and **JUnit 5/Mockito** tests.
- Containerized each service with multi-stage **Docker** builds, configured the EC2 server with **Ansible**, and ran the stack with **Docker Compose** behind an **Nginx** reverse proxy; added a **GitHub Actions** CI workflow.

**User Management System** **Oct 2024 – Dec 2024**

Spring Boot · Hibernate/JPA · MySQL | GitHub

- Built a CRUD REST API with dynamic search and filtering, request validation, and centralized exception handling.
- Modeled a normalized 6-table MySQL schema with JPA entity mappings; tested and documented all endpoints in Postman.

## EDUCATION

**Sri Venkateswara College of Engineering**, Tirupati **2022 – May 2026**

B.Tech, Electronics & Communication Engineering CGPA: 8.5 / 10

## CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate — Oracle (Sep 2025) · Problem Solving, Intermediate — HackerRank (Apr 2026) · SQL, Intermediate — HackerRank (Mar 2026) · Java Programming — GeeksforGeeks (Mar 2026)

## ACHIEVEMENTS

- Solved **100+ Data Structures & Algorithms** problems on LeetCode; cleared HackerRank Problem Solving (Intermediate).
- Built a Spring Boot LMS backend at **Design Spark Challenge 2026**, a 48-hour innovation marathon (RTIH, Andhra Pradesh).